

Natdemy AI CCTV Project

Project Brief

One camera platform that understands what is happening in a space and turns it into an alert, a report or an ERP entry, customised for each customer.

Status: attendance demo working, office testing in progress.

Already proven: camera, face recognition, person identified, attendance marked. Attendance is the first use case, not the product.

How every solution works

Six steps, the same for every customer. We build them once and reuse them.

Camera	AI	Event	Rule	Action	App / ERP
Sees the space	Detects people and objects	Something happened	Does it matter?	Alert or record	Where it lands

Example: face recognised, employee identified, attendance rule, attendance marked, ERP updated.

Where we can apply it

Use case	What the AI checks	Example trigger	Action
Attendance	Face recognition	Employee or student enters	Attendance marked, ERP updated
Pool safety	Person, zone, movement	Child enters pool area after hours	Alert to security
Retail analytics	Person detection, tracking	Queue grows between 5 and 6 PM	Analytics dashboard
Industrial safety	Zones, PPE, falls	Worker in restricted area without PPE	Alert, safety record
School / institute	Faces, zones, crowd	Visitor enters restricted area	Attendance, alert
Farm monitoring	Person and animal detection	Person crosses boundary at night	Alert to owner
Security / intrusion	Unknown person, zones	Unknown person in restricted area at 2 AM	Alert to security
Visitor management	Detection, face recognition	Visitor enters or exits	Visitor record

Positioning for safety cases: we do not promise drowning detection. We promise continuous monitoring and an early alert when something unusual happens.

Core capabilities to build once

Area	Capabilities
People	Count, entry and exit, face recognition, dwell time, movement, presence in a zone
Objects and activity	Machine usage, door usage, room occupancy, parking and vehicle movement
Appearance	Uniform, PPE, approximate clothing colour

Test for every capability: what useful problem does it solve? We do not collect data only because AI can detect it.

Testing ground: our office camera

Test	What we measure
Attendance	Recognition accuracy, repeated marking prevented
Entry / exit	How accurately people entering and leaving are counted
Coffee machine	Number of users, usage time, peak period
Vending machine	Number of users, time spent, peak activity
Meeting room	Occupied or empty, approximate occupancy, duration
Clothing	Main colour, uniform detection (technology test only)

Roadmap

#	Step	Scope
1	Stabilise attendance	Make the current face recognition and attendance system reliable
2	Test in the office	Use the installed IP camera to learn what is practically possible
3	Build reusable components	Person detection, face recognition, tracking, counting, zones, dwell time, basic events, alerts
4	Connect to applications and ERP	Send useful events into business systems
5	Package customer solutions	Turn the strongest use cases into customer-ready offerings

Long-term direction

One AI CCTV platform, many solutions. Customers use different parts of the same base: camera, AI, rules, application and ERP. Attendance, access and security, education, retail, industrial safety, pool and residential safety, farm monitoring and visitor management all sit on it.

Current goal

Not a finished commercial product yet. The question for now: what can we make the camera understand, and what useful action can we trigger from it? Attendance proved the idea. Next we prove it works on different real-world problems.